

# Amit Kumar Mohanty

mohantyamit2003@gmail.com | +91 9354937256 | github.com/amitmohanty022 | linkedin.com/in/amitkrmohanty

Target Role: Platform Engineer I - Generative AI Engineer (Fresher)

## PROFESSIONAL SUMMARY

Generative AI Engineer and Computer Science graduate (AI/ML specialization) with strong Python fundamentals (NumPy, Pandas) and a public GitHub portfolio of functioning GenAI projects, including an LLM-powered assistant and an AI chatbot. Solid theoretical foundation in NLP, tokenization, embeddings, and Transformer architecture. Hands-on with LLM-powered applications, smart chatbots, and data pipelines for AI training and retrieval using LangChain, LlamaIndex, and vector databases (FAISS, ChromaDB, Pinecone). Comfortable with Git/GitHub collaboration and eager to turn academic and project experience into real-world enterprise AI solutions.

## EDUCATION

### K.R. Mangalam University, Gurgaon

July 2021 - July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML)

### Birla Institute of Technology and Science, Pilani (BITS Pilani)

July 2026 - July 2028

M.Tech in Artificial Intelligence and Machine Learning (Pursuing)

## EXPERIENCE

### Research Associate - AI, Keywords Studios India - Gurgaon, Haryana

May 2025 - Present

- Trained and optimized LLM-powered GenAI agents that autonomously execute multi-step workflows, improving task-completion reliability.
- Optimized data pipelines for AI training and retrieval, delivering 85% of self-generated training datasets to production using Python (NumPy, Pandas).
- Awarded "Rookie of the Year 2025" for exceeding all KPIs; collaborated with a global team using Git and GitHub.

### Artificial Intelligence Intern, Infosys - Remote

Nov 2024 - Feb 2025

- Built a smart AI chatbot / call-center assistant with LangChain, boosting response accuracy by 35% by integrating LLMs into Python services.
- Applied prompt engineering and Transformer-based NLP to reduce hallucinations by 20% and improve conversation quality.
- Deployed LLM features that reduced system latency by 25% within a 12-engineer collaborative team.

## PROJECTS

### Dynamic Screen Companion - LLM Assistant (github.com/amitmohanty022)

June 2025 - Present

- Built a real-time GenAI assistant integrating LLM APIs (Gemini/OpenAI) to analyze live screen activity and deliver conversational voice responses.
- Engineered a Python FastAPI backend and an embeddings + retrieval pipeline (tokenization, embeddings, vector search) with 95%+ extraction accuracy.

### Diabetic Optiscan - Vision Transformer (Peer-reviewed research)

Jan 2025 - June 2025

- Built a Vision Transformer (ViT-B/16) on the APTOS dataset to detect diabetic retinopathy; co-authored a peer-reviewed paper.
- Designed a custom wavelet technique that boosted detection precision by 50% over legacy methods.

### Currency Detection App for the Visually Impaired

July 2024 - Dec 2024

- Trained an image-classification model on 200,000+ banknote images (96% accuracy) and deployed it in a mobile app with voice feedback.

## SKILLS

**Programming:** Python (NumPy, Pandas), SQL, C++, R, Git, GitHub

**NLP & Transformers:** NLP, Tokenization, Embeddings, Transformer Architecture, LLMs, Hugging Face, BERT, Prompt Engineering, Fine-tuning (LoRA/QLoRA)

**GenAI Frameworks:** LangChain, LangGraph, LlamaIndex, RAG, Agent Workflows, OpenAI API, Gemini API

**Vector Databases & Data:** FAISS, ChromaDB, Pinecone, Vector Search, Data Pipelines, MySQL, MongoDB

**ML & Deployment:** PyTorch, TensorFlow, Scikit-learn, OpenCV, FastAPI, Docker, AWS, GCP (Vertex AI)

## CERTIFICATIONS

- Ducat India - Data Science Professional Training (Python, ML, statistics, predictive modeling), Apr 2024 - Apr 2025.
- ISRO - AI/ML for Geodata Analysis (applied ML/DL to remote-sensing imagery).
- Portfolio: functioning GenAI projects (LLM assistant, AI chatbot, document analysis) on github.com/amitmohanty022.